

AP1
AP2
HACK THE NORTH 2026 · CUSTOM ESP32-C3 FIRMWARE

Badge Finder

Finding a friend with wifi

A

(you)

AP3

B



THE SCENARIO

You're trying to find your friend at Hack the North.

1,000+

hackers, one building

0 bars

of signal inside

“by the tables”

describes the entire venue

We flashed a custom firmware



your friend is in here. somewhere.

THE HARDWARE

The badge in your hands

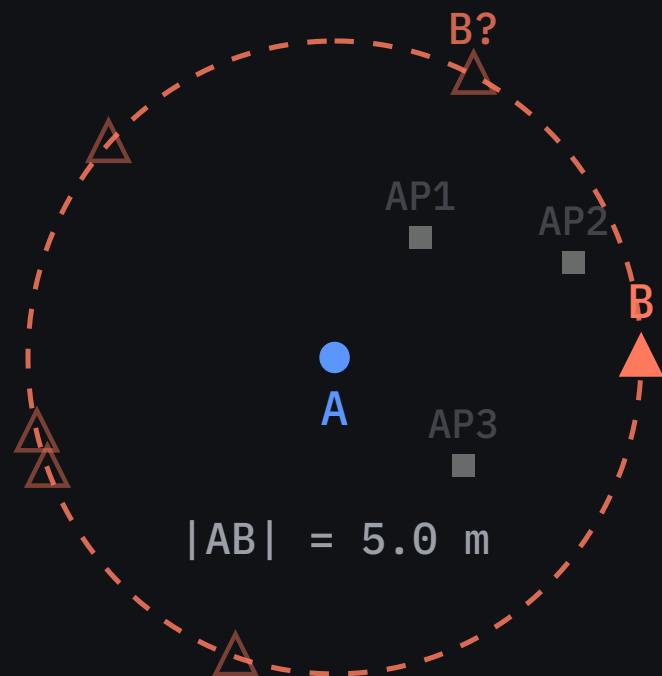
WHAT IT HAS

- WiFi scan, ESP-NOW, SoftAP
- 802.11mc FTM ranging
- BLE advertise + scan
- Accelerometer
- 6 RGB LEDs, 320×240 display

WHAT IT DOESN'T

- GPS
- Magnetometer
- Gyroscope
- UWB / AoA antennas

It has no idea which way it is pointing.



5.0 m — in every direction at once

WHAT WE CAN MEASURE

A distance is only a ring

FTM gives the distance.

Only a radius — B is anywhere on it.

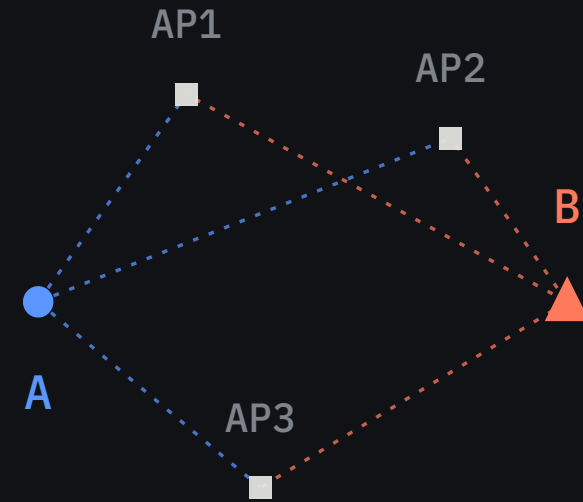
We need a direction.

THE KEY INSIGHT

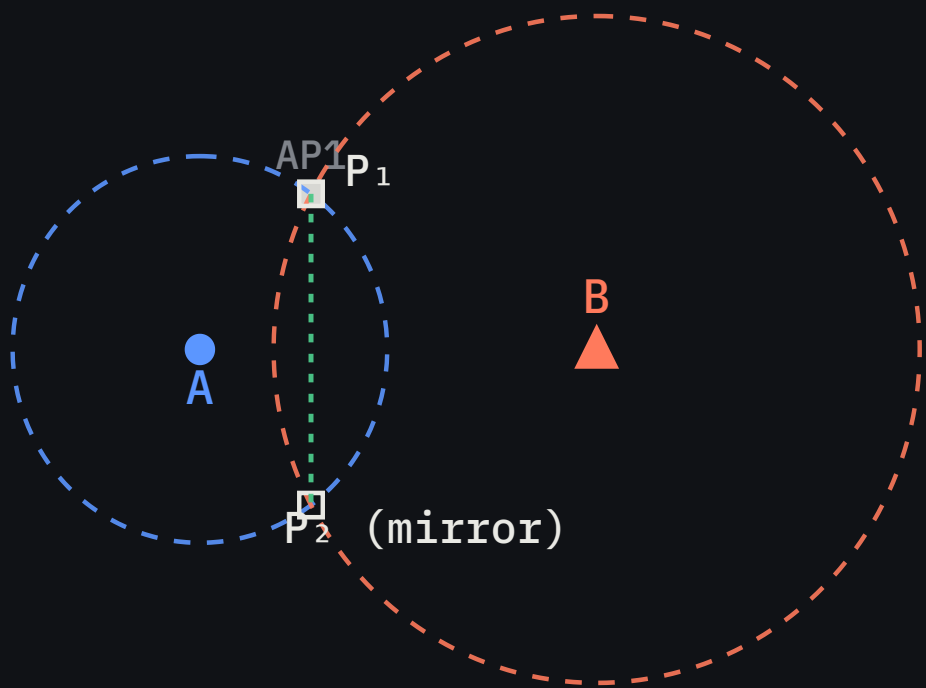
We are not alone in the room

5–8 APs. Never move. **Both badges see them.**

Free, static reference points.



B replies over ESP-NOW — 114 bytes



one AP → two candidates

STAGE 1 · FIRST CAPTURE

Two circles, two answers

RSSI → rough distances. The circles cross at **two mirror points**.

per AP · A at (0,0), B at (d,0)

$$x = (d^2 + dA^2 - dB^2) / 2d$$

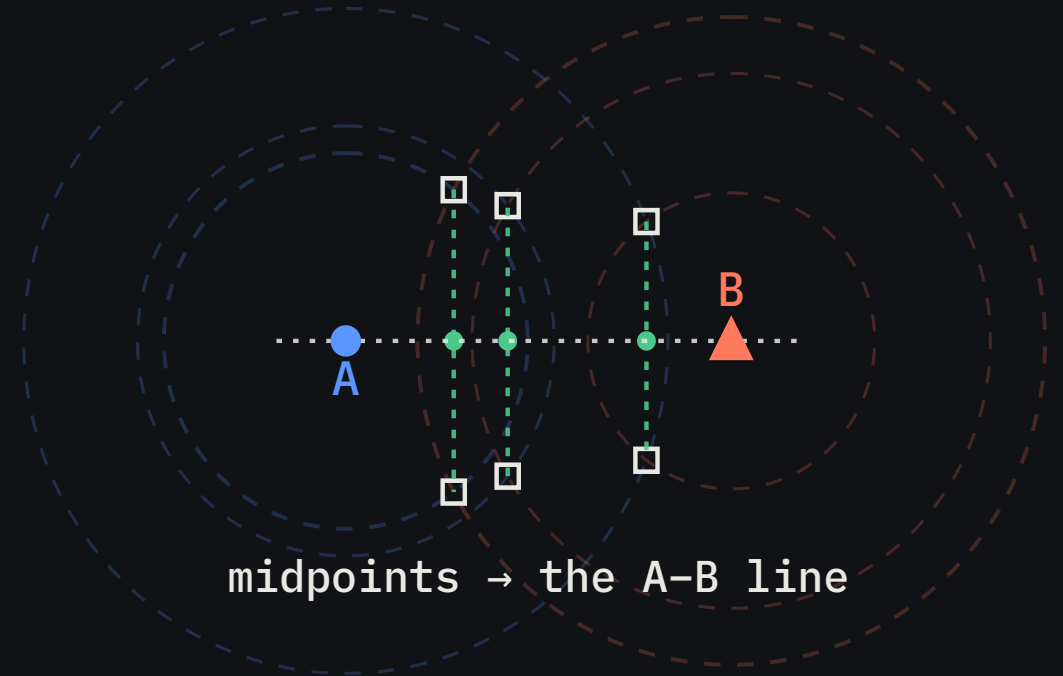
$$y = \pm\sqrt{(dA^2 - x^2)}$$

STAGE 1 · GEOMETRY CHECK

The chords hand us the A–B line

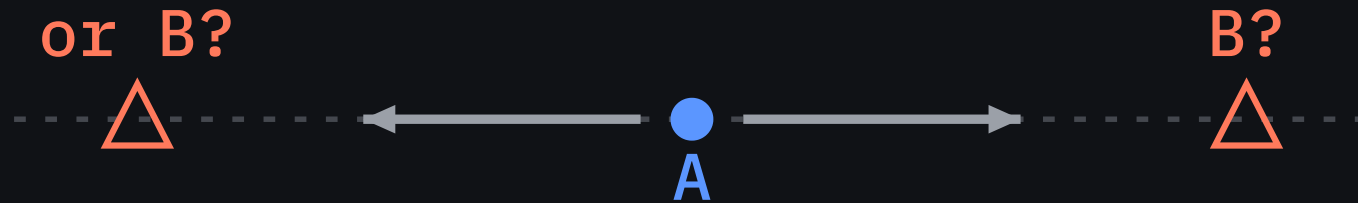
The chords are **all parallel** — 90.00°.

Their midpoints give **the A–B line**.

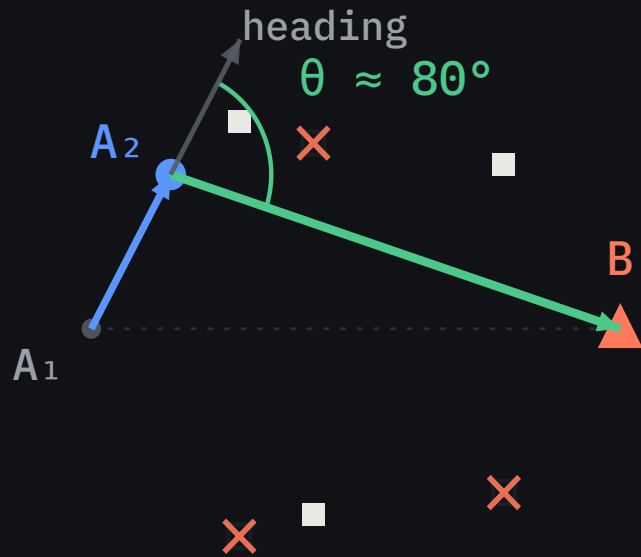


STAGE 1 · THE LIMIT

We know the axis — not the direction



Both directions fit — toward B, or away.



A₁, B, A₂ pin the map to your walk

STAGE 2 · SECOND CAPTURE

Walk a few steps, measure again

Steps count the metres. Direction = **straight ahead**.

Re-scan solves B — **up to a mirror**. **The spin picks the side.**

on-badge solver · walk = +x

$$x = (d_1^2 - d_2^2 + s^2) / 2s$$

$$\theta = \text{atan2}(\sqrt{d_1^2 - x^2}, x)$$

A_2 (you)

B — found



DEMO

live, on two badges

ERRORS & ASSUMPTIONS

What can go wrong

THE ERROR

30–50%

per-scan WiFi distance error

Fix: a rolling average. The noise moves; the truth doesn't.

THE ASSUMPTIONS

- No bodies blocking (10–20 dB)
- Light obstruction only
- You walk straight
- Left or right: walk one way — warmer wins

Questions?

Hack the North 2026 · custom ESP-IDF firmware